Recap.

① For every RV X, $\phi_X(t) = E(e^{itX}) = \begin{cases} \sum_x e^{itx} p_X(x) \\ \int_{-\infty}^{\infty} e^{itx} p_X(x)\,dx \end{cases}$, where i is imaginary unit.

② Every characteristic function uniquely matches to a distribution.
For RVs X and Y, given $X \perp\!\!\!\perp Y$, then $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$.

③ Markov's inequality: for RV X defined on $[0,+\infty)$ and $a>0$, $P(X \geq a) \leq \frac{E(X)}{a}$.
Chebyshev's inequality: for any RV X and $b>0$, $P(|X - E(X)| \geq b) \leq \frac{Var(X)}{b^2}$.

④ "Given a sample of independent and identically distributed value $X_1, X_2 \cdots X_n$, average value of sample $\bar{X}_n = \frac{X_1+X_2+\cdots+X_n}{n}$ converges to expected value $E(X_i) = \mu$."
wlln (converges in probability): $\bar{X}_n \xrightarrow{P} \mu$, i.e. for $\forall \varepsilon > 0$, $P(|\bar{X}_n - \mu| > \varepsilon) \to 0$ when $n \to \infty$.
slln (almost surely converges): $\bar{X}_n \xrightarrow{a.s.} \mu$, i.e. $P(\lim_{n\to\infty} \bar{X}_n = \mu) = 1$.

Exercises.

Ex 1. Hint: Taylor series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$.

For $Z \sim Poi(\lambda)$,
$$\phi_Z(t) = \sum_{n=0}^{\infty} e^{itn} \cdot P(Z=n) = \sum_{n=0}^{\infty} e^{itn} \frac{e^{-\lambda}\lambda^n}{n!} = e^{-\lambda} \sum_{n=0}^{\infty} \frac{(e^{it}\lambda)^n}{n!} = e^{-\lambda} e^{e^{it}\lambda}$$

$\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t) = e^{(e^{it}-1)\lambda_1} \cdot e^{(e^{it}-1)\lambda_2} = e^{(e^{it}-1)(\lambda_1+\lambda_2)}$, $X+Y \sim Poi(\lambda_1+\lambda_2)$.

Ex 2. Hint: For discrete RV mapped to N and $n \in N$, $P(X > n) = P(X \geq n+1)$.

(i) $E(N_T) = E(N_A) + E(N_B) = 150$. Since $N_A \perp\!\!\!\perp N_B$, $Var(N_T) = Var(N_A) + Var(N_B) = 150$.

(ii) With Chebyshev's inequality: $P(|N_T - 150| \geq b) \leq \frac{150}{b^2}$.
$P(N_T > 200) = P(N_T \geq 201) = P(N_T - 150 \geq 201 - 150) \leq P(|N_T - 150| \geq 51) \leq \frac{150}{51^2} \approx 5.77\%$.

(iii) With Markov's inequality: $P(N_T > 200) = P(N_T \geq 201) \leq \frac{150}{201} = 74.63\%$.
"Variance contains more information. Chebyshev's inequality gives a tighter bound."

(iv) $N_A \sim Poi(50)$, $N_B \sim Poi(100)$, since Expectation = Variance.

(v) $N_T = N_A + N_B \sim Poi(150)$, $P(N_T > 200) = 1 - P(N_T \leq 200) \approx 0.0042\%$.
"Inequalities are estimations without known distribution, not precise."

Ex 3. Hint: $\chi_{\bar{D}^2} = \begin{cases} 1 & \text{if } (x,y) \in \bar{D}^2 \\ 0 & \text{otherwise} \end{cases}$

(i) $p_X(x) = \int_{-\infty}^{\infty} p_{X,Y}(x,y)\,dy = \begin{cases} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi}\,dy = \frac{2}{\pi}\sqrt{1-x^2} & \text{if } x \in [-1,1] \\ 0 & \text{otherwise} \end{cases}$

$p_Y(y) = \begin{cases} \frac{2}{\pi}\sqrt{1-y^2} & \text{if } y \in [-1,1] \\ 0 & \text{otherwise} \end{cases}$

(ii) $E(X) = \int_{-\infty}^{\infty} x \cdot p_X(x)\,dx = \int_{-1}^{1} \frac{2x}{\pi}\sqrt{1-x^2}\,dx = 0.$ ($\frac{2x}{\pi}$ is odd function and $\sqrt{1-x^2}$ is even)

$E(Y) = 0.$

$$\begin{aligned}
Var(X) &= E(X^2) - (E(X))^2 \\
&= E(X^2) \\
&= \int_{-\infty}^{\infty} x^2 p_X(x)\,dx \\
&= \frac{2}{\pi}\int_{-1}^{1} x^2\sqrt{1-x^2}\,dx \\
&= \frac{2}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2(u)\cos(u)\cos(u)\,du \\
&= \frac{2}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{4}(\sin 2u)^2\,du \\
&= \frac{2}{\pi}\times\frac{1}{4}\times 2\int_{0}^{\frac{\pi}{2}} (\sin 2u)^2\,du \\
&= \frac{1}{\pi}\int_{0}^{\frac{\pi}{2}} \frac{1-\cos 4u}{2}\,du \\
&= \frac{1}{\pi}\left[\frac{u}{2} - 2\sin(4u)\right]_0^{\frac{\pi}{2}} \\
&= \frac{1}{4}.
\end{aligned}$$

(Let $x = \sin(u)$, then $x' = \sin'(u)$, $1 = \cos(u)$, $\frac{dx}{du} = \cos(u)$)

($2\sin u\cos u = \sin 2u$)

($(\sin 2u)^2$ is even function)

($(\sin u)^2 = \frac{1-\cos 2u}{2}$)

$Var(Y) = \frac{1}{4}.$

(iii) ① Verify $p_{X,Y}(x,y) \overset{?}{=} p_X(x)\cdot p_Y(y)$.

Pick $(x,y) = (0.8, 0.8)$, $p_X(x) = p_Y(y) > 0 = p_{X,Y}(x,y)$. not independent.

② Verify $cov(X,Y) = E(X\cdot Y) - E(X)E(Y) = E(X\cdot Y) \overset{?}{=} 0.$

$$\begin{aligned}
E(X\cdot Y) &= \iint_{\bar{D}^2} x y \frac{1}{\pi}\,dx\,dy \\
&= \frac{1}{\pi}\iint_{\bar{D}^2} r^2\cos\theta\sin\theta\,dx\,dy \\
&= \frac{1}{\pi}\int_0^{2\pi}\int_0^1 r^2\cos\theta\sin\theta\, r\,dr\,d\theta \\
&= \frac{1}{\pi}\int_0^{2\pi}\cos\theta\sin\theta\,d\theta\int_0^1 r^3\,dr \\
&= \frac{1}{\pi}\int_0^{2\pi}\frac{1}{2}\sin(2\theta)\,d\theta\int_0^1 r^3\,dr \\
&= 0,\quad \text{uncorrelated.}
\end{aligned}$$

(define $x = r\cos\theta$, $y = r\sin\theta$ with $r \in [0,1]$, $\theta \in [0, 2\pi)$)

($dx\,dy = |\det(J)|\,dr\,d\theta = \left|\begin{pmatrix} \frac{dx}{dr} & \frac{dx}{d\theta} \\ \frac{dy}{dr} & \frac{dy}{d\theta} \end{pmatrix}\right| dr\,d\theta = r\,dr\,d\theta$)

($\int_0^{2\pi}\sin(2\theta)\,d\theta = 0$)

Ex 4. Solution shown in the code form.